UniMamba: A Unified Cross-Modal Mamba Framework for Remote Sensing Semantic Segmentation

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***Abstract*—Fusing optical imagery with complementary modalities (X-modality), such as light detection and ranging (LiDAR) and synthetic aperture radar (SAR), is essential for robust semantic segmentation in complex environments. Although recent modality-agnostic models improve generalizability beyond fixed-pair methods, they still lack a unified and efficient architecture to process diverse combinations, including RGB-X, multispectral (MSI-X), and hyperspectral (HSI-X). To address this, we propose UniMamba, a unified cross-modal fusion Mamba (FuseMa) framework for general multimodal semantic segmentation. Specifically, UniMamba employs a unified encoder that processes diverse 2-D–3-D inputs without modality-specific modifications. The encoder integrates HyperMamba blocks to jointly model spatial, spectral, and frequency features while capturing global context with linear complexity. Each encoding stage further incorporates a cross HyperMamba (CroHMa) module for explicit cross-modal interaction, seamlessly followed by a FuseMa module for complementary feature fusion. Extensive experiments on three benchmarks demonstrate that UniMamba outperforms state-of-the-art methods and generalizes robustly across diverse multimodal settings. Code is available at https://github.com/xumzhang.**

***Index Terms*—Classification, cross-modality fusion, deep learning (DL), Mamba, remote sensing, semantic segmentation.**

I. Introduction

SEMANTIC segmentation assigns semantic labels to individual pixels, enabling fine-grained land-cover interpretation from aerial and satellite imagery [1], [2]. Although optical imagery is the primary data source, its quality often degrades under adverse conditions, such as low illumination, atmospheric interference, and smoke [3]. Integrating complementary modalities, including digital elevation models (DEMs), synthetic aperture radar (SAR), and light detection and ranging (LiDAR), enriches feature representation and improves robustness. However, effectively aligning and fusing heterogeneous sensor data remains a major challenge.

To address this challenge, various strategies have been developed, which are generally categorized into pixel-level [4], feature-level [5], and decision-level fusion [6]. Among these, feature-level fusion is widely preferred due to its scalability and effectiveness [7]. Early feature-level methods relied on traditional statistical algorithms, and their dependence on hand-crafted features restricts representation capacity and generalizability.

Recent advances in deep learning (DL) have substantially improved feature representation for multimodal semantic segmentation. Early DL approaches [8], [9], [10], [11] primarily adopted convolutional neural networks (CNNs) [12], [13] because of their effectiveness in extracting hierarchical features and capturing local spatial patterns. Nevertheless, their limited receptive fields hinder global context modeling, while fixed convolutional kernels and shared weights reduce adaptability to spatial heterogeneity and image degradation [3], [14].

Transformers [15], [16] overcome these limitations through dynamic self-attention, enabling global context modeling. However, their quadratic complexity with respect to input size incurs substantial computational and memory costs, particularly for high-resolution remote sensing imagery. Hybrid CNN-Transformer architectures [7], [17], [18], [19], [20] alleviate these issues, but remain computationally expensive. Consequently, efficiently modeling long-range dependencies in large-scale multimodal data remains an open challenge.

Recently, Mamba [21], a selective structured state-space model (SSM), has emerged as a compelling alternative for vision tasks, offering dynamic weight generation and global dependency modeling with linear complexity [14]. Its parallel scanning algorithm [22] accelerates token processing, improving computational efficiency on modern hardware like GPUs. Moreover, Mamba's selective mechanism and hardware-aware design further enhance training and inference performance [23]. These advantages enable it to outperform

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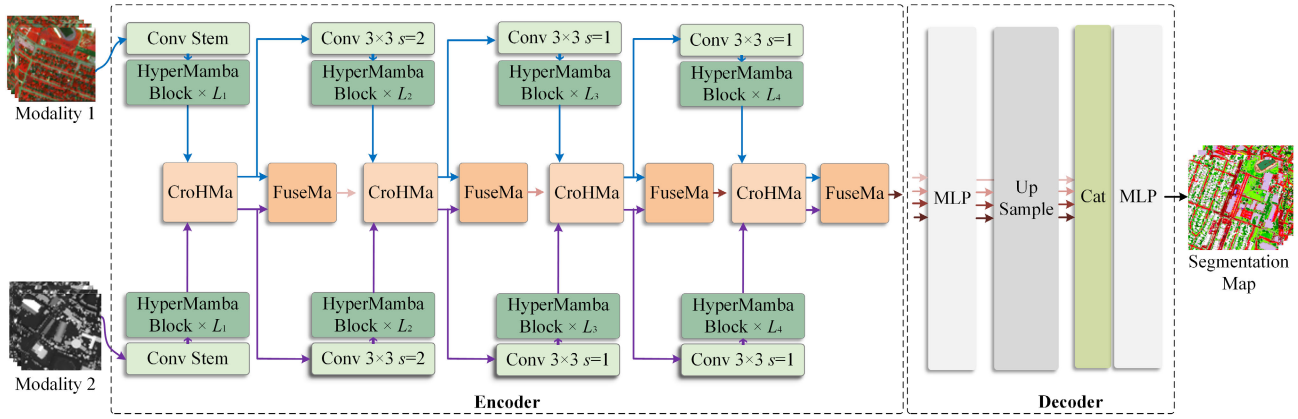


Fig. 1. Overall architecture of the proposed UniMamba. Here, "Cat" represents the concatenation operation.

Transformers in long-sequence modeling tasks, such as large-scale image processing. Variants like Vision Mamba [24], VMamba [25], and Sigma [3] have demonstrated strong performance in vision tasks, including image classification [24], semantic segmentation [3], [26], and object detection [27], and multimodal fusion.

Despite these advances, existing methods still exhibit three main limitations. First, most approaches are designed for fixed modality pairs (e.g., HSI-LiDAR [20], [28], [29]). Although some modality-agnostic architectures support RGB-X [30] or HSI-X segmentation [7], where X denotes a complementary modality (e.g., SAR or LiDAR), they cannot generalize across both settings. Second, optical imagery, particularly HSI and MSI, often suffers from spectral redundancy, noise, and inconsistent interband contrast caused by different imaging mechanisms. Although frequency-domain representations yield robust features to mitigate these issues, existing vision Mamba models [24], [25] extract primarily spatial features, neglecting the critical spectral and frequency information. Third, existing Mamba-based fusion modules [3], [23], [31] lack a unified mechanism for explicit cross-modal interaction across diverse modality combinations, limiting their generalizability.

To address these limitations, we propose UniMamba, a unified cross-modal fusion Mamba (FuseMa) framework (see Fig. 1). UniMamba consists of a dual-backbone encoder and a lightweight all-MLP decoder. The encoder comprises four stages, each integrating HyperMamba blocks to jointly model spatial, spectral, and frequency-domain information. Each stage further incorporates a cross HyperMamba (CroHMa) module for explicit cross-modal interaction, followed by a FuseMa module for complementary feature fusion. The decoder aggregates multistage fused features to generate the final segmentation map.

The main contributions can be summarized as follows.

1) Proposing UniMamba, a unified framework for multimodal semantic segmentation that supports RGB-X, MSI-X, and HSI-X within a single architecture. To the best of the authors' knowledge, UniMamba is the first unified framework supporting these multimodal settings.
2) Developing a HyperMamba block to jointly model spatial, spectral, and frequency-domain representations for modality-adaptive feature learning from heterogeneous multimodal inputs.
3) Designing CroHMa and FuseMa modules for explicit cross-modal interaction and complementary feature fusion, respectively, with linear complexity, enabling efficient multimodal integration.

The remainder of this article is organized as follows. Section II reviews related work on multimodal semantic segmentation and SSMs. Section III details the proposed UniMamba framework, while Section IV evaluates its performance. Section V presents ablation studies on HyperMamba, CroHMa, and FuseMa, followed by a discussion. Finally, Section VI concludes this study and suggests future research directions.

II. Related Works

A. Multimodal Semantic Segmentation

Multimodal semantic segmentation exploits complementary information from different modalities to improve scene understanding. For example, RGB provides rich visual appearance, HSI captures spectral information, DEM encodes elevation information, and LiDAR provides 3-D geometric structure [7].

Early approaches typically employ two-stream CNNs, where each stream processes one modality independently before feature fusion [8], [32]. Feature fusion was commonly performed using elementwise addition [9], multiplication [33], or concatenation [8]. To improve feature representation and cross-modal fusion, subsequent studies introduced attention mechanisms [18], [30], [34], [35]. More recently, Transformer-based methods [15], [17], [36] have achieved superior performance by modeling long-range dependencies and global context [20], [28], [29]. For example, the local information interaction Transformer (LIIT) [28] employs a dual-branch Transformer to model interactions between HSI and LiDAR. TokenFusion [37] introduces a unified Transformer framework for homogeneous and heterogeneous modality fusion. Fusion_HCT [20] employs a hybrid CNN-Transformer architecture to capture local and global features in sequence.

Despite these advances, most existing methods remain tailored to specific modality combinations [20], [28], [38]. For example, FusionHCT [20] and LIIT [28] focus on HSI-LiDAR segmentation, while MSFMamba [39] and MCA-Mamba [40] are designed for HSI-LiDAR/SAR classification. To improve generalization, several modality-agnostic frameworks have been developed. CMX [30] and RSGMamba [41] target RGB-X segmentation, while local-to-global cross-

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modal attention-aware fusion (LoGoCAF) [7] and CoMiX [42] focus on HSI-X segmentation. However, none provides a unified solution for RGB-X, MSI-X, and HSI-X semantic segmentation. UniMamba addresses this gap through a unified modality-agnostic design.

Furthermore, cross-modal interaction is fundamental to multimodal semantic segmentation. Existing methods primarily employ cross attention to exploit complementary information across modalities [7], [20], [30]. Although effective, cross attention requires explicit pairwise correlation modeling, resulting in high computational complexity. In contrast, CroHMa performs cross-modal interaction within a Mamba-based framework, enabling efficient long-range dependency modeling with linear complexity.

B. State-Space Models

SSMs [43], [44], originating from classical control theory, excel at modeling long-range dependencies. The structured state-space sequence (S4) model [44] established the foundation for modern state-space architectures.

Building on S4, Mamba [21] introduces a selective state-space mechanism that dynamically captures input-relevant information, together with a hardware-aware algorithm that accelerates training and inference. It achieves superior throughput, lower latency, and better overall performance than similarly sized Transformers. These advances have facilitated its extension beyond natural language processing to computer vision. Vision Mamba [24] extends Mamba with bidirectional scanning to capture spatial relationships, while VMamba [25] further enhances spatial modeling with cross scanning.

Sigma [3] extends Mamba to RGB-X semantic segmentation using a Siamese encoder and a channel-aware decoder. Its encoder introduces the cross Mamba block (CroMB) and concat Mamba block (ConMB) for cross-modal interaction and feature fusion, respectively. However, existing Mamba-based methods primarily model spatial representations and are designed for specific modality combinations. In contrast, UniMamba jointly models spatial, spectral, and frequency-domain representations within a unified framework that supports RGB-X, MSI-X, and HSI-X semantic segmentation.

III. Methodology

A. Overall Architecture

As illustrated in Fig. 1, UniMamba comprises a dual-branch encoder and an ALL-MLP decoder [45]. Each encoder branch begins with a stem layer consisting of a 1×1 convolution, batch normalization, and SiLU activation to align the channel dimensions. The resulting features are progressively encoded through four stages. Each stage includes a 3×3 convolution for channel adjustment and local feature extraction, followed by a stack of HyperMamba blocks. To optimize the computation-performance tradeoff, $2\times$ downsampling is performed only in the second stage using a stride-2 convolution. After each stage, CroHMa and FuseMa are cascaded to drive cross-modal interaction and feature fusion, respectively. Finally, the multilevel fused features are aggregated by the ALL-MLP decoder to generate the segmentation mask. The HyperMamba block, CroHMa, FuseMa, and ALL-MLP decoder are described in Sections III-B–III-E, respectively.

B. HyperMamba Block

The original Mamba block is designed for 1-D sequence modeling, limiting its spatial awareness. Vision Mamba [24] and VMamba [25] improve spatial modeling through bidirectional and cross-scanning mechanisms, respectively. However, these methods are not tailored to remote sensing imagery, which features rich spectral information, ambiguous boundaries, and modality-specific noise. Moreover, existing Mamba variants inevitably degrade local spatial structures during 1-D serialization and underutilize spectral and frequency-domain information.

To address these limitations, we propose the HyperMamba block [see Fig. 2(a)], formulated as

$$\mathbf{X}' = \text{HyperMamba}(\text{LN}(\mathbf{X})) + \mathbf{X} \tag{1}$$

$$\mathbf{Y} = \text{DWFFN}((\text{LN}(\mathbf{X}')) + \mathbf{X}' \tag{2}$$

where $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ denotes the input feature map, with $H \times W$ and C representing the spatial resolution and channel dimension, respectively. LN is layer normalization [46], while $\mathbf{X}'$ and $\mathbf{Y}$ are the intermediate and output features, respectively. HyperMamba [see Fig. 2(b)] is expressed as

$$\text{HyperMamba} = f\left(\text{LN}\left(\text{SS4D}\left(\text{SiLU}\left(\text{DWConv}\left(f\left(\mathbf{X}\right)\right)\right)\right)\right)\right) \tag{3}$$

where f denotes a linear layer. The depthwise feed-forward network (DWFFN) is defined as

$$\text{DWFFN} = \text{Conv}_{1\times1}\left(\text{GELU}\left(\text{DWConv}\left(\text{Conv}_{1\times1}\left(\mathbf{X}\right)\right)\right)\right) \tag{4}$$

where $\text{Conv}_{1\times1}$ denotes a 1×1 convolution. The first $\text{Conv}_{1\times1}$ doubles the channel dimension, while the second restores it. The depthwise convolution (DWConv) enhances local spatial modeling with limited computational overhead. By combining long-range dependency modeling with local spatial refinement, the HyperMamba block captures both global context and local details, making it well suited for dense prediction in remote sensing imagery.

The 4-D-selective-scan (SS4D) performs selective scanning along three directions, combined with Fourier domain modeling. As shown in Fig. 2(c), the input is first flattened into three sequences corresponding to forward spatial, reverse spatial, and channelwise interaction. Each sequence is independently processed by the S6 module [25] to capture direction-specific long-range dependencies, while the forward spatial sequence is additionally transformed into the Fourier domain for frequency learning. The output of SS4D integrates all components

$$\mathbf{Y}_{\text{SS4D}} = \mathbf{Y}_{\text{spa-fwd}} + \mathbf{Y}_{\text{spa-rev}} + \mathbf{Y}_{\text{spe}} + \mathbf{Y}_{\text{f}} \tag{5}$$

where $\mathbf{Y}_{\text{spa-fwd}}$, $\mathbf{Y}_{\text{spa-rev}}$, and $\mathbf{Y}_{\text{spe}}$ are the outputs of the S6 module applied to the forward spatial, reverse spatial, and channel sequences, respectively, and $\mathbf{Y}_{\text{f}}$ is the output from the Fourier-domain modeling. Details on spectral–spatial sequence modeling and frequency-domain modeling are presented below.

1) Spectral–Spatial Sequence Modeling: Given an input tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, we define the forward spatial sequence $\mathbf{X}_{\text{spa-fwd}}$, reverse spatial sequence $\mathbf{X}_{\text{spa-rev}}$, and channel interaction sequence $\mathbf{X}_{\text{spe}}$ as

$$\mathbf{X}_{\text{spa-fwd}} = \left[X^1_{i,1}, \ldots, X^1_{i,HW}, \ldots, X^C_{i,1}, \ldots, X^C_{i,HW}\right] \tag{6}$$

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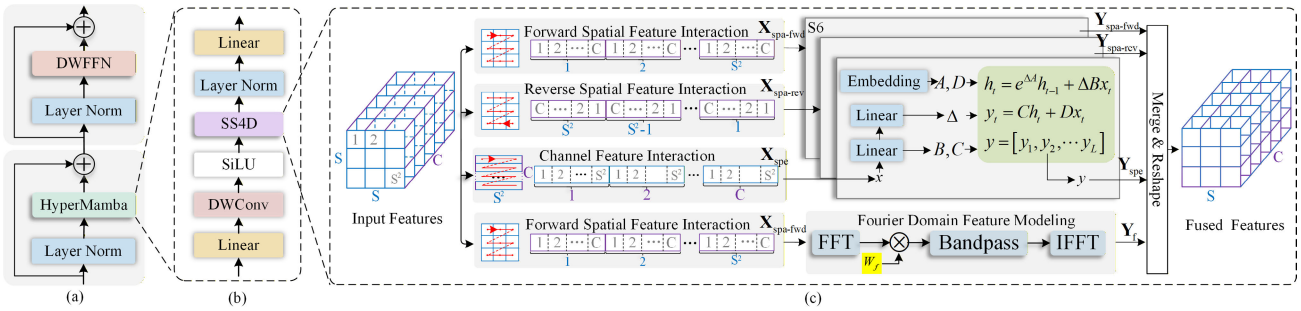


Fig. 2. Illustration of the proposed HyperMamba block. (a) HyperMamba block. (b) HyperMamba. (c) SS4D.

$$\mathbf{X}_{\text{spa-rev}} = \Phi_{\text{flip}}\left(\mathbf{X}_{\text{spa-fwd}}\right) \tag{7}$$

$$\mathbf{X}_{\text{spe}} = \left[X^1_{i,1}, \ldots, X^C_{i,1}, \ldots, X^1_{i,HW}, \ldots, X^C_{i,HW}\right] \tag{8}$$

where Φ_{flip} denotes the flipping function. Each sequence is processed independently using the S6 module

$$\begin{aligned}\mathbf{Y}_{\text{spa-fwd}} &= \Phi_{\text{S6}}\left(\mathbf{X}_{\text{spa-fwd}}\right)\\ \mathbf{Y}_{\text{spa-rev}} &= \Phi_{\text{S6}}\left(\mathbf{X}_{\text{spa-rev}}\right)\\ \mathbf{Y}_{\text{spe}} &= \Phi_{\text{S6}}\left(\mathbf{X}_{\text{spe}}\right)\end{aligned} \tag{9}$$

where Φ_{S6} denotes the S6 module. The resulting sequences are reshaped to their original tensor layouts.

This three-direction design captures bidirectional spatial context and spectral dependencies with limited computational overhead. Additional directions, such as reverse spectral scanning, increase computation and redundancy while providing only marginal gains, as verified in Section V-A.

2) Frequency-Domain Modeling: Frequency components provide vital complementary cues: high frequencies capture edges and textures, whereas low frequencies represent global structures and smooth variations [47]. UniMamba exploits these properties by transforming the forward spatial sequence into the frequency domain via the fast Fourier transform (FFT). The frequency representations are then processed by a learnable 1-D convolution and subjected to stage-specific frequency filtering, before being reconstructed via the inverse FFT (IFFT)

$$\mathbf{Y}_{\text{f}} = \text{IFFT}\left(\text{Bandpass}\left(W_f\left(\text{FFT}\left(\mathbf{X}_{\text{spa-fwd}}\right)\right)\right)\right) \tag{10}$$

where $W_{\text{f}} \in \mathbb{R}^{C\times C\times k}$ denotes the weights of a 1-D convolution with kernel size $k = 5$.

To align frequency extraction with the hierarchical nature of the encoder, Bandpass(·) dynamically applies high-pass, bandpass, or low-pass filtering depending on the network stage

$$\text{Bandpass}\left(\mathbf{X}\right) = \begin{cases}\mathbf{X}\left(|\mathbf{X}| < S_{\text{low}}\right), & \mathbf{X} \in F_{\text{low}}\\ \mathbf{X}\left(S_{\text{low}} < |\mathbf{X}| < S_{\text{high}}\right), & \mathbf{X} \in F_{\text{mid}}\\ \mathbf{X}\left(|\mathbf{X}| > S_{\text{high}}\right), & \mathbf{X} \in F_{\text{high}}\end{cases} \tag{11}$$

where F_{low}, F_{med}, and F_{high} represent features from the low level (Stage 1), mid level (Stages 2 and 3), and high level (Stage 4) of UniMamba, respectively. The filtering thresholds are empirically set to $S_{\text{low}} = 0.2$ and $S_{\text{high}} = 0.8$ in our experiments. This stage-specific frequency strategy emphasizes the frequency components that are most relevant to each representation level.

Notably, by serializing inputs into sequences, SS4D can process 1-D, 2-D, and 3-D data without architectural modification. This enables HyperMamba to accommodate diverse modalities without modality-specific designs.

C. Cross HyperMamba

In multimodal remote sensing, modality discrepancies often lead to feature misalignment. To mitigate this issue, we propose the CroHMa module for cross-modal feature interaction.

As shown in Fig. 3(a), CroHMa enhances multimodal feature interactions across spatial, spectral, and frequency domains. Given input features $\mathbf{I}_1, \mathbf{I}_2 \in \mathbb{R}^{HWD\times C}$, CroHMa linearly transforms them into two distinct feature spaces

$$\mathbf{I}^{\text{x}}_i,\ \mathbf{I}^{\text{z}}_i = f\left(\text{LN}\left(\mathbf{I}_i\right)\right),\ f\left(\text{LN}\left(\mathbf{I}_i\right)\right), \quad i = 1,\ 2. \tag{12}$$

Similar to SS4D, $\mathbf{I}^{\text{x}}_i$ is flattened into forward spatial sequences ($\mathbf{I}^{\text{spa-fwd}}_i$), reverse spatial sequences ($\mathbf{I}^{\text{spa-rev}}_i$), and channel sequences ($\mathbf{I}^{\text{spe}}_i$), respectively, where $i = 1,\ 2$. Each sequence is then independently processed by the S6 model

$$\begin{aligned}\mathbf{Y}^{\text{spa-fwd}}_i &= \Phi_{\text{S6}}\left(\mathbf{I}^{\text{spa-fwd}}_i\right)\\ \mathbf{Y}^{\text{spa-rev}}_i &= \Phi_{\text{S6}}\left(\mathbf{I}^{\text{spa-rev}}_i\right)\\ \mathbf{Y}^{\text{spe}}_i &= \Phi_{\text{S6}}\left(\mathbf{I}^{\text{spe}}_i\right), \quad i = 1,\ 2.\end{aligned} \tag{13}$$

Meanwhile, the forward spatial sequences are transformed into the Fourier domain to extract frequency information

$$\mathbf{Y}^{f}_i = \text{IFFT}(\text{Bandpass}(W_f\left(\text{FFT}\left(\mathbf{I}^{\text{spa-fwd}}_i\right)\right), \quad i = 1,\ 2. \tag{14}$$

Inspired by cross-attention mechanisms [7], [30] widely used for cross-modal fusion, CroHMa uses the activation $\mathbf{I}^{\text{z}}_i$ of one modality to modulate the representations of the other. Specifically, we apply a SiLU activation to generate cross-modal attention weights

$$\mathbf{Y}^{\text{z}}_i = \text{SiLU}\left(\mathbf{I}^{\text{z}}_i\right), \quad i = 1,\ 2. \tag{15}$$

These activations modulate the spatial, spectral, and frequency features of the counterpart modality via elementwise multiplication

$$\mathbf{Y}_1 = \mathbf{Y}^{\text{z}}_2 \cdot \mathbf{Y}^{\text{spa-fwd}}_1 + \mathbf{Y}^{\text{z}}_2 \cdot \mathbf{Y}^{\text{spa-rev}}_1 + \mathbf{Y}^{\text{z}}_2 \cdot \mathbf{Y}^{\text{spe}}_1 + \mathbf{Y}^{\text{z}}_2 \cdot \mathbf{Y}^{\text{f}}_1 \tag{16}$$

$$\mathbf{Y}_2 = \mathbf{Y}^{\text{z}}_1 \cdot \mathbf{Y}^{\text{spa-fwd}}_2 + \mathbf{Y}^{\text{z}}_1 \cdot \mathbf{Y}^{\text{spa-rev}}_2 + \mathbf{Y}^{\text{z}}_1 \cdot \mathbf{Y}^{\text{spe}}_2 + \mathbf{Y}^{\text{z}}_1 \cdot \mathbf{Y}^{\text{f}}_2. \tag{17}$$

The enhanced features $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are linearly projected and fused with their corresponding inputs through residual connections

$$\mathbf{Y}^{\text{out}}_i = f\left(\mathbf{Y}_i\right) + \mathbf{I}_i, \quad 1,\ 2. \tag{18}$$

Through this design, CroHMa provides an efficient alternative to cross attention for cross-modal interaction. Unlike cross

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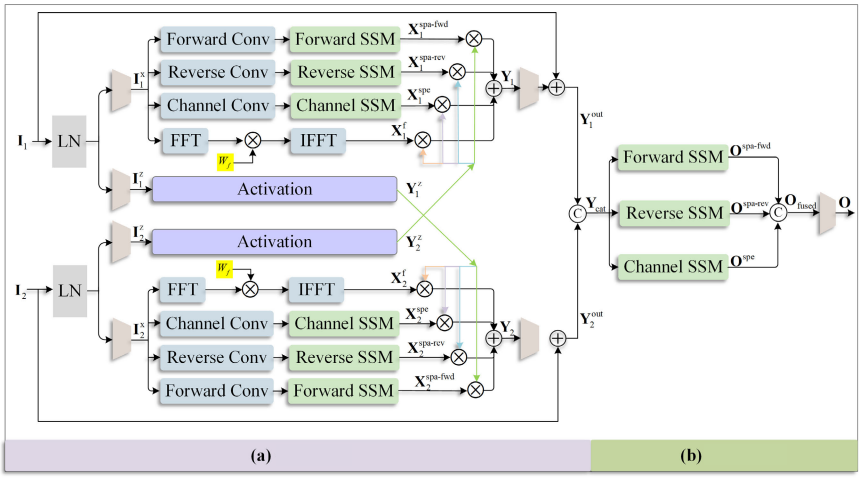


Fig. 3. Illustration of the proposed (a) CroHMa and (b) FuseMa modules.

attention, which computes pairwise correlations with quadratic complexity, CroHMa employs a selective state-space mechanism with linear complexity. Regarding feature alignment, cross attention explicitly models token-to-token correspondences. In contrast, CroHMa achieves implicit alignment by globally contextualizing features via state-space modeling, reducing sensitivity to local spatial offsets. However, under severe geometric misalignment, the lack of explicit alignment may reduce performance due to the bounded receptive field of state-space modeling.

D. Fusion Mamba

In FuseMa [see Fig. 3(b)], the outputs $\mathbf{Y}_1^{\text{out}}$ and $\mathbf{Y}_2^{\text{out}}$ from CroHMa are first concatenated

$$\mathbf{Y}_{\text{cat}} = \left[\mathbf{Y}_1^{\text{out}}, \mathbf{Y}_2^{\text{out}}\right]. \tag{19}$$

$\mathbf{Y}_{\text{cat}}$ is then reshaped into forward spatial ($\mathbf{Y}_{\text{cat}}^{\text{spa-fwd}}$), reverse spatial ($\mathbf{Y}_{\text{cat}}^{\text{spa-rev}}$), and spectral sequences ($\mathbf{Y}_{\text{cat}}^{\text{spe}}$), which are independently processed by the S6 module

$$\begin{aligned}\mathbf{O}^{\text{spa-fwd}} &= \Phi_{\text{S6}}\left(\mathbf{Y}_{\text{cat}}^{\text{spa-fwd}}\right)\\ \mathbf{O}^{\text{spa-rev}} &= \Phi_{\text{S6}}\left(\mathbf{Y}_{\text{cat}}^{\text{spa-rev}}\right)\\ \mathbf{O}^{\text{spe}} &= \Phi_{\text{S6}}\left(\mathbf{Y}_{\text{cat}}^{\text{spe}}\right).\end{aligned} \tag{20}$$

Finally, the three outputs are concatenated and projected to obtain the fused representation

$$\mathbf{O} = f\left(\left[\mathbf{O}^{\text{spa-fwd}}, \mathbf{O}^{\text{spa-rev}}, \mathbf{O}^{\text{spe}}\right]\right) \tag{21}$$

where f denotes the projection layer.

E. All-MLP Decoder

Decoders, such as the UNet decoder [12] and the channel-aware Mamba decoder [3], often introduce substantial computational overhead. Given the strong feature representations produced by the encoder, we adopt the lightweight ALL-MLP decoder [45].

As shown in Fig. 1, the fused features from each encoder stage are projected by separate MLP layers to align their channel dimensions and then upsampled to the input resolution. The resulting features are concatenated and processed by two additional MLP layers for feature fusion and prediction, respectively. This lightweight design avoids computationally intensive decoding operations and improves the efficiency of UniMamba.

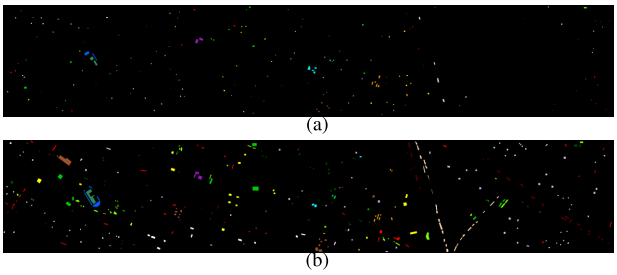


Fig. 4. DFC2013 dataset: spatial distribution of (a) training samples and (b) test samples.

TABLE I
LAND-COVER TYPES AND THE NUMBER OF TRAINING AND TEST SAMPLES ON THE DFC2013 DATASET

ID	Color	Land-cover Type	Training	Test
C1		Healthy grass	198	1053
C2		Stressed grass	190	1064
C3		Synthetic grass	192	505
C4		Trees	188	1056
C5		Soil	186	1056
C6		Water	182	143
C7		Residential	196	1072
C8		Commercial	191	1053
C9		Road	193	1059
C10		Highway	191	1036
C11		Railway	181	1054
C12		Parking Lot 1	192	1041
C13		Parking Lot 2	184	285
C14		Tennis Court	181	247
C15		Running Track	187	473

IV. EXPERIMENTS AND RESULTS

A. Description of Datasets

We evaluated UniMamba on three datasets: DFC2013, DFC2018, and Hunan, which combine HSI-LiDAR, RGB-MS_LiDAR, and MSI-SAR modalities, respectively.

1) DFC2013 Dataset: This dataset, acquired in June 2012, consists of an HSI and a LiDAR-derived digital surface model

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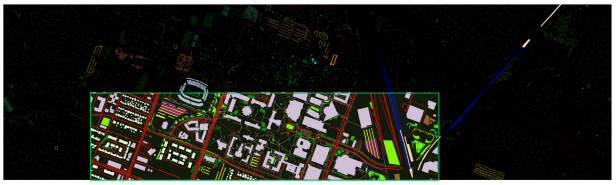


Fig. 5. DFC2018 dataset: spatial distribution of training (green box) and test (outside the green box) samples.

TABLE II
LAND-COVER TYPES AND THE NUMBER OF TRAINING AND TEST SAMPLES ON THE DFC2018 DATASET

ID	Color	Land-cover Type	Training	Test
C1		Healthy grass	39196	20000
C2		Stressed grass	130008	20000
C3		Artificial turf	2736	20000
C4		Evergreen trees	54322	20000
C5		Deciduous trees	20172	20000
C6		Bare earth	18064	20000
C7		Water	1064	1628
C8		Residential buildings	158995	20000
C9		Non-residential buildings	894769	20000
C10		Roads	183283	20000
C11		Sidewalks	136035	20000
C12		Crosswalks	6059	5345
C13		Major thoroughfares	185438	20000
C14		Highways	39438	20000
C15		Railways	27748	11232
C16		Paved parking lots	45932	20000
C17		Unpaved parking lots	587	3524
C18		Cars	26289	20000
C19		Trains	21479	20000
C20		Stadium seats	27296	20000

(DSM). Both share the same spatial dimensions (349 × 1905) and spatial resolution (2.5 m). The HSI contains 144 spectral bands covering wavelengths from 380 to 1050 nm. The dataset contains 15 classes. Their training and test sample distributions are presented in Table I and Fig. 4, based on the official data split [48]. Before the experiment, the entire image was divided into 128 × 128 subimages with 50% overlap to reduce memory requirements, resulting in 180 subimages.

2) DFC2018 Dataset: This dataset was acquired in February 2017 over the University of Houston and surrounding areas. It comprises: 1) multispectral LiDAR (MS-LiDAR) point cloud data at wavelengths of 1550, 1064, and 532 nm, along with DSM and DEM data at a 50-cm ground sampling distance (GSD); 2) 48-band HSI spanning 380–1050 nm at a 1-m GSD; and 3) high-resolution RGB imagery at a 5-cm GSD. The ground truth (GT) contains 20 urban land-cover classes at a 0.5-m GSD and has dimensions of 8344 × 2404 pixels.

The normalized DSM (NDSM), calculated as NDSM = DSM − DEM, represents the actual elevation of the objects. To evaluate the performance of RGB-X data in our approach, experiments were conducted using RGB and MS-LiDAR data. Before the experiment, the entire image was divided into 128 × 128 subimages with 50% overlap, resulting in 1425 subimages for training and 4978 for testing. The numbers of training and test samples and their spatial distributions are shown in Table II and Fig. 5, respectively.

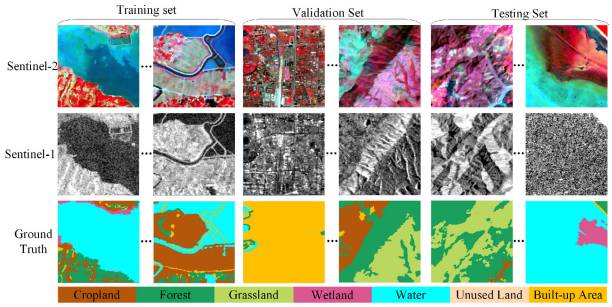


Fig. 6. Hunan dataset.

3) Hunan Dataset: This dataset, including Sentinel-2 (MSI), Sentinel-1 (SAR), and SRTM digital elevation data (SRTM), was acquired in 2017 over Hunan, China [49]. We used Sentinel-1 and Sentinel-2 data for our experiments. For Sentinel-2, ten bands were used, excluding B1, B9, and B10, because of their limited relevance to land-cover classification. The dataset contains 400 training, 50 validation, and 50 test images, each with a spatial size of 256 × 256 pixels. It defines 7 classes, as shown in Fig. 6. More details are available in [49].

B. Experimental Settings and Evaluation Metrics

1) Implementation Details: All models were implemented using the PyTorch framework and executed on an NVIDIA GTX 3090 GPU. All datasets were normalized to [0, 1] before the experiments.

The parameters of UniMamba were optimized using the Adam optimizer with a cross-entropy loss function. UniMamba was trained for 500 epochs with a weight decay of 0.01. The initial learning rate was set to 6×10^{-5} and adjusted using a poly-learning rate schedule. According to the input sizes, the batch size was set to 4 for the DFC2013 and DFC2018 datasets and 2 for Hunan dataset. For the comparison methods, publicly available source codes were used. To ensure fair comparisons, the same data preprocessing strategy was applied to all methods. Results are averaged over five runs for objectivity.

2) Evaluation Measures: Performance was evaluated using intersection over union (IoU), mean IoU (mIoU), producer's accuracy (PA), average PA (AA), and overall accuracy (OA).

C. Comparison With Other Methods

We compared UniMamba with several multimodal models, including coupled adversarial-learning-based classification (CALC) [9], multimodal fusion Transformer (MFT) [50], multistage information complementary fusion network based on flexible-mixup (Flex-MCFNet) [51], LoGoCAF [7], CMX [30], and Sigma [3]. These models are designed for different modality combinations: CALC and MFT are patch-based classification networks for HSI and LiDAR data; Flex-MCFNet and LoGoCAF are tailored for HSI-X semantic segmentation; while CMX and Sigma focus on RGB-X segmentation tasks. Specifically, CALC integrates HSI and LiDAR features using a combination of CNNs and attention mechanisms, while MFT employs a ViT-based architecture to process HSI patches

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TABLE III

CLASSIFICATION ACCURACY ON THE DFC2013 DATASET. THE BEST VALUES ARE MARKED IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

	HSI-LiDAR				HSI-X				RGB-X				ALL	
Method	CACL		MFT		Flex-MCFNet		LoGoCAF		CMX		Sigma		UniMamba	
Metrics	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA
C1	81.10	80.63	81.72	82.67	**85.21**	**85.77**	83.02	83.10	84.05	84.23	82.78	83.10	83.10	83.10
C2	82.73	80.83	83.76	84.48	84.05	87.40	85.06	85.06	85.07	85.15	78.22	78.29	**87.67**	**95.58**
C3	59.32	85.55	88.77	98.68	99.60	99.86	**99.61**	**100.00**	75.04	99.41	99.21	**100.00**	95.64	**100.00**
C4	96.25	91.00	91.53	92.33	93.83	94.37	83.88	**100.00**	92.23	92.23	87.56	93.94	**97.78**	**100.00**
C5	**100.00**	99.34	95.96	98.22	99.89	99.92	**100.00**	**100.00**	99.91	99.91	99.81	99.81	**100.00**	**100.00**
C6	97.92	93.01	90.59	95.45	54.17	**100.00**	67.45	**100.00**	37.27	99.30	34.71	**100.00**	**100.00**	**100.00**
C7	69.75	83.12	66.29	85.11	84.60	93.22	85.65	86.29	62.75	90.21	83.89	91.32	**95.06**	**95.06**
C8	74.67	85.28	68.70	78.73	79.05	**93.81**	78.30	83.95	74.18	79.11	75.48	92.97	**87.10**	89.74
C9	85.02	86.03	84.43	87.73	87.88	95.53	80.32	80.55	77.00	81.87	89.40	93.20	**93.51**	**98.02**
C10	58.90	60.91	57.48	60.94	70.37	71.39	72.66	94.40	59.69	63.61	65.11	65.73	**91.68**	**100.00**
C11	73.74	94.88	64.59	96.01	83.75	83.81	85.60	99.24	78.87	83.59	79.70	99.81	**86.96**	**100.00**
C12	72.82	88.38	84.87	92.88	75.35	96.76	89.59	**100.00**	73.06	98.46	85.68	95.97	**95.41**	97.89
C13	43.61	92.98	86.73	90.01	80.68	**95.74**	**87.72**	87.72	87.17	92.98	80.40	84.91	73.33	73.33
C14	99.60	99.60	96.05	99.87	**100.00**	**100.00**	**100.00**	**100.00**	99.20	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
C15	99.58	**100.00**	97.71	96.09	99.58	**100.00**	99.37	99.58	99.37	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Average	79.67	88.10	83.02	89.28	85.20	93.17	86.55	93.33	78.99	90.00	82.80	91.94	**92.48**	**95.51**
OA	86.24		87.76		91.76		92.11		87.49		90.52		**95.86**	

TABLE IV

CLASSIFICATION ACCURACY ON THE DFC2018 DATASET. THE BEST VALUES ARE MARKED IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

	HSI-LiDAR				HSI-X				RGB-X				ALL	
Method	CACL		MFT		Flex-MCFNet		LoGoCAF		CMX		Sigma		UniMamba	
Metrics	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA
C1	73.51	91.15	79.87	90.30	73.97	92.93	80.30	89.13	71.83	83.59	**80.81**	89.73	76.83	**94.20**
C2	54.81	**82.24**	53.43	76.55	61.77	75.03	59.75	80.41	62.35	85.69	**67.22**	79.47	56.45	75.77
C3	69.53	71.81	79.47	83.05	71.91	75.29	81.37	82.95	88.02	89.24	**95.66**	**97.62**	84.54	86.53
C4	61.10	91.86	**77.12**	84.98	69.30	93.55	67.03	89.13	76.62	93.15	69.99	**94.05**	75.76	93.80
C5	19.47	19.62	46.76	48.16	48.22	**49.16**	32.35	32.79	**48.60**	48.89	47.02	48.16	45.82	46.31
C6	14.73	14.74	31.59	33.03	35.62	35.99	32.94	33.17	59.22	61.23	57.44	63.39	**72.91**	**73.49**
C7	2.18	5.96	3.03	8.66	3.26	8.97	3.29	8.05	0.00	0.00	0.00	0.00	**5.12**	**10.75**
C8	82.78	**96.27**	78.18	95.19	79.62	91.78	82.33	94.05	82.74	93.34	82.07	90.74	**85.79**	94.18
C9	43.41	94.36	45.13	94.75	54.30	95.22	55.03	94.77	60.05	94.56	**63.89**	**95.44**	63.54	94.28
C10	31.91	81.56	29.30	78.58	33.42	84.16	33.26	81.82	32.41	87.04	32.22	85.74	**44.50**	**87.25**
C11	44.81	77.85	51.89	76.29	47.39	74.25	49.49	75.41	52.99	**79.67**	**55.99**	77.12	55.39	75.99
C12	8.86	19.81	8.94	11.77	14.73	26.51	14.36	**33.90**	10.17	12.18	1.84	2.04	**22.84**	26.16
C13	24.10	38.51	20.84	34.89	19.79	34.70	**25.66**	39.33	16.44	27.41	21.70	39.68	24.42	**49.89**
C14	**19.22**	19.64	18.94	**20.02**	17.54	18.42	18.80	19.25	17.94	18.43	18.53	18.99	18.95	19.09
C15	89.59	94.31	86.72	98.85	85.02	99.69	79.58	99.95	88.73	98.59	90.17	99.73	**98.28**	**100.00**
C16	62.26	74.49	59.11	70.93	58.91	70.06	68.40	83.65	68.54	81.56	66.01	77.96	**71.68**	**85.93**
C17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C18	42.94	43.16	48.13	48.59	51.51	55.03	54.41	56.69	48.69	50.73	46.12	47.26	**60.60**	**62.26**
C19	81.10	81.36	84.68	85.39	77.51	77.72	93.99	95.08	**99.28**	**99.74**	98.09	98.56	93.79	94.03
C20	42.21	43.64	74.63	75.56	78.35	79.58	77.34	79.23	80.53	80.56	**90.24**	**90.42**	83.32	85.42
Average	43.43	57.12	48.89	60.77	49.11	61.90	50.49	63.44	53.26	64.28	54.25	64.80	**57.03**	**67.76**
OA	63.26		67.63		68.28		69.80		72.19		73.21		**75.05**	

along with LiDAR class tokens. Flex-MCFNet and LoGoCAF combine CNNs and attention mechanisms for HSI-X classification. Meanwhile, CMX, a Transformer-based framework, and Sigma, a Mamba-based network, are designed for RGB-X semantic segmentation.

1) Quantitative Results and Analysis: The quantitative results in Tables III–V show that UniMamba outperforms the competing methods across all three datasets, achieving the highest mIoU, AA, and OA. These outcomes highlight its robustness and adaptability in multimodal segmentation tasks.

CALC and MFT, tailored to HSI-LiDAR classification, achieve moderate performance on the DFC2013 dataset. Although CALC benefits from CNN-based feature extraction and attention mechanisms, it struggles to capture the complex interactions between HSI and LiDAR data, limiting its accuracy. MFT outperforms CALC in capturing complex data patterns; however, both are constrained by patch-based operations that inherently limit long-range dependency modeling.

Flex-MCFNet and LoGoCAF perform competitively across all datasets. Flex-MCFNet employs adaptive spectral–spatial

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TABLE V
CLASSIFICATION ACCURACY ON HUNAN DATASET. THE BEST VALUES ARE MARKED IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

	HSI-LiDAR				HSI-X				RGB-X				ALL	
Method	CACL		MFT		Flex-MCFNet		LoGoCAF		CMX		Sigma		UniMamba	
Metrics	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA	IoU	PA
C1	65.73	76.79	65.15	80.19	67.19	80.01	69.05	80.46	66.66	80.55	67.56	**81.94**	**69.32**	81.33
C2	81.33	**96.05**	82.15	94.50	81.94	93.08	82.00	94.44	81.82	92.33	82.33	93.25	**83.26**	92.61
C3	17.41	22.11	18.01	22.08	23.28	30.45	13.83	16.35	26.96	36.70	22.94	28.16	**30.53**	**39.26**
C4	14.86	42.23	18.19	41.48	17.65	40.61	17.00	37.86	15.79	40.44	**20.17**	39.57	17.89	**43.56**
C5	73.49	80.71	77.33	85.77	76.81	85.06	76.26	86.04	75.01	82.46	**79.02**	88.01	75.22	84.58
C6	19.21	21.89	18.01	20.27	20.29	23.47	**29.77**	**37.11**	23.54	28.15	21.04	24.57	27.33	33.38
C7	65.12	75.79	63.86	72.66	66.02	79.12	**66.82**	79.12	65.45	77.62	65.78	76.56	66.68	**80.50**
Average	48.16	59.37	48.96	59.56	50.46	61.68	50.68	61.62	50.75	62.61	51.26	61.72	**52.89**	**65.03**
OA	81.32		82.07		82.50		82.83		82.02		83.17		**83.26**	

feature fusion modules and performs well on DFC2013 and DFC2018 datasets. Nonetheless, its patch-based classification strategy limits performance in complex scenes, such as those on Hunan dataset. In contrast, LoGoCAF, adopting a conventional segmentation framework, enables broader contextual modeling and slightly outperforms Flex-MCFNet in such cases.

CMX and Sigma deliver strong results on the DFC2018 and Hunan datasets. Sigma, in particular, achieves second-best performance in mIoU, AA, and OA across both datasets, highlighting its robustness and adaptability to RGB-MS_LiDAR and MSI-SAR scenarios. However, both methods suffer a notable performance decline on the Houston dataset, primarily due to their limited capacity for rich spectral extraction in HSIs. Similarly, although Flex-MCFNet and MFT perform reasonably well on the Houston dataset, their effectiveness drops considerably in MSI–SAR tasks.

These findings suggest that models specialized for specific modality combinations often struggle to generalize across diverse multimodal settings. In contrast, UniMamba integrates HSI-X, RGB-X, and MSI-X within a unified framework and consistently outperforms the competing methods, demonstrating stronger adaptability to heterogeneous multimodal remote sensing tasks.

Nevertheless, UniMamba achieves lower accuracy than some competing methods on several challenging categories, including C1 (healthy grass) and C13 (parking lot 2) of DFC2013, as well as C7 (water) and C17 (unpaved parking lots) of DFC2018. Distinguishing C1 and C13 from their highly similar counterparts (C2 and C12) relies heavily on fine-grained spectral responses, localized microtextures, and spatial gradients. CNN-based methods may capture these cues more effectively through local inductive biases, whereas UniMamba's global state-space modeling is less sensitive to localized high-frequency details. Furthermore, misclassifications on C7 and C17 are largely due to shadows and irregular textures. Although the frequency-domain branch provides complementary texture and structural information, its class-agnostic filtering does not explicitly enlarge interclass feature margins. Therefore, incorporating interclass contrastive learning within the frequency domain represents a promising avenue for future work to resolve these class ambiguities.

2) Qualitative Results and Analysis: Classification maps, false color images, and GT are shown in Figs. 7–9, with black boxes highlighting challenging or distinctive regions.

Across the three datasets, UniMamba consistently produces smoother and more accurate segmentation maps than competing methods. It excels in boundary delineation and class separation, significantly reducing misclassifications in complex regions, as illustrated in the black-box areas. For example, on the DFC2018 dataset, UniMamba accurately distinguishes cars from paved parking lots, even within densely built-up zones.

In contrast, CALC, MFT, and Flex-MCFNet rely on patch-based classification, which limits their ability to capture long-range dependencies. This shortcoming is especially evident in scenes requiring broader contextual understanding. For instance, CALC struggles with boundary preservation [see Fig. 8(c)], whereas MFT improves boundary clarity but suffers from fragmented segmentation in heterogeneous regions, exemplified in wetlands on Hunan dataset. Similarly, Flex-MCFNet exhibits boundary artifacts on the Houston dataset, particularly in regions with gradual spectral transitions. Its performance also declines on the DFC2018 dataset, where long-range contextual information is essential, such as that required for major thoroughfares.

LoGoCAF, CMX, and Sigma produce smoother and more coherent outputs but underperform in complex land cover regions. LoGoCAF struggles to distinguish trains from railways on the DFC2018 dataset. Although CMX and Sigma perform better in such scenarios, their performance decreases on spectrally rich datasets like Houston. As shown in Fig. 7(f), Sigma confuses spectrally similar classes (e.g., highways and commercial areas), indicating difficulty in capturing subtle spectral variations and leading to increased misclassification in ambiguous areas.

Overall, UniMamba demonstrates strong robustness and adaptability by effectively integrating spectral, spatial, and structural features. Its accurate classification across diverse datasets highlights its versatility for multimodal segmentation tasks, making it well-suited for real-world applications.

3) Efficiency Analysis: We evaluated the efficiency of UniMamba by comparing it with other methods in terms of the number of parameters (Params), floating point operations

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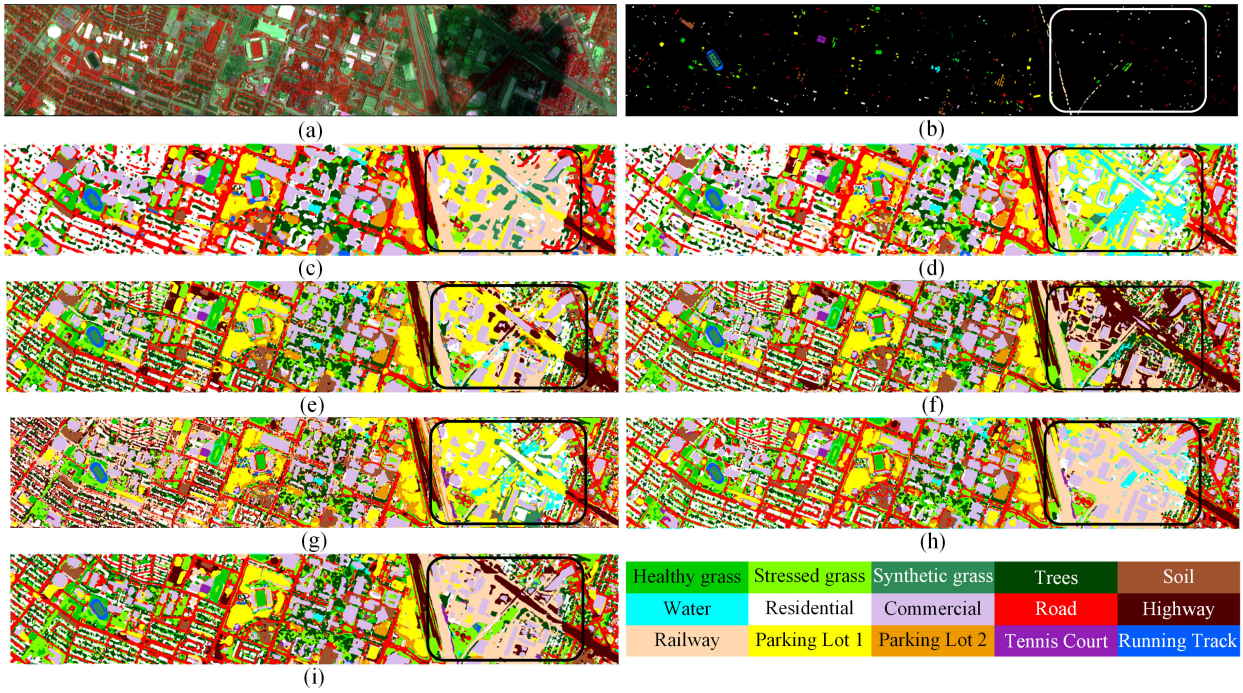


Fig. 7. Visual comparison on the DFC2013 dataset. (a) False color image. (b) GT. (c) CALC. (d) MFT. (e) Flex-MCFNet. (f) LoGoCAF. (g) CMX. (h) Sigma. (i) UniMamba.

TABLE VI

COMPARISON OF THE NUMBER OF PARAMS, FLOPS, TRAINING (TRAIN), AND INFERENCE (INFER) TIME OF DIFFERENT METHODS ON THE DFC2013 DATASET

Method	Params (M)	FLOPs (G)	Train (s)	Infer (s)
CACL	0.34	1.84	1696.08	54.21
MFT	0.31	1.62	19010.97	44.79
Flex-MCFNet	7.14	83.81	7859.25	64.21
LoGoCAF	7.75	93.68	9444.23	22.63
CMX	7.89	81.67	8282.43	21.69
Sigma	7.18	102.56	33797.297	28.04
UniMamba	6.41	90.46	20526.22	13.16

(FLOPs), training time (Train), and inference time (Infer) on the DFC2013 dataset.

As summarized in Table VI, UniMamba achieves a favorable tradeoff between computational complexity and accuracy, with 6.41M Params and 90.46G FLOPs—moderate relative to Flex-MCFNet (83.81G FLOPs) and Sigma (102.56G FLOPs). Moreover, UniMamba achieves near real-time inference (13.16 s), outperforming LoGoCAF (22.63 s) and Flex-MCFNet (64.21 s). Its efficiency benefits from Mamba's linear-complexity modeling of long-range dependencies, avoiding the quadratic cost of self-attention and the redundant computations inherent in patch-based methods.

Although UniMamba requires longer training than LoGoCAF and CMX, this cost is offset by substantial gains in accuracy and robustness, primarily attributable to CroHMa and FuseMa, which enable more effective cross-modal fusion.

4) Comparison With Recent Mamba-Based Methods: Table VII compares UniMamba with recent Mamba-based methods, including RSGMamba [41], MSFMamba [39], MCAMamba [40], and CoMiX [42]. These methods perform competitively within their target modalities but remain constrained by modality-specific designs. RSGMamba targets RGB-X segmentation and shows lower performance on the HSI-LiDAR DFC2013 dataset. In contrast, MSFMamba, MCAMamba, and CoMiX focus on HSI-based tasks and exhibit reduced performance on the RGB-X DFC2018 and MSI-X Hunan datasets. UniMamba achieves the best overall performance across all three datasets, demonstrating the effectiveness of its unified modality-agnostic design.

V. DISCUSSION

A. Ablation Study

We conducted ablation studies to evaluate the contribution of the core components in UniMamba. The baseline consisted of an encoder composed of standard Vision Mamba blocks and an ALL-MLP decoder.

1) Effectiveness of HyperMamba, CroHMa, and FuseMa: To evaluate the contribution of HyperMamba, CroHMa, and FuseMa, we replace HyperMamba with standard Vision Mamba blocks and remove CroHMa or FuseMa individually. Without CroHMa, the two modalities are processed independently. Without FuseMa, the dual-branch features are fused by averaging.

As shown in Table VIII, replacing Vision Mamba with HyperMamba consistently improves mIoU across all datasets, demonstrating its effectiveness in feature modeling. Incorporating CroHMa or FuseMa further boosts mIoU and OA, indicating their critical roles in bridging modality gaps.

Crucially, these substantial accuracy gains are achieved with high cost-effectiveness. As detailed in the complexity metrics (Params and FLOPs) on DFC2013, integrating HyperMamba, CroHMa, and FuseMa introduces only moderate computational overhead. Ultimately, integrating all three components, UniMamba achieves the best performance across all datasets. These results confirm that UniMamba's superiority stems

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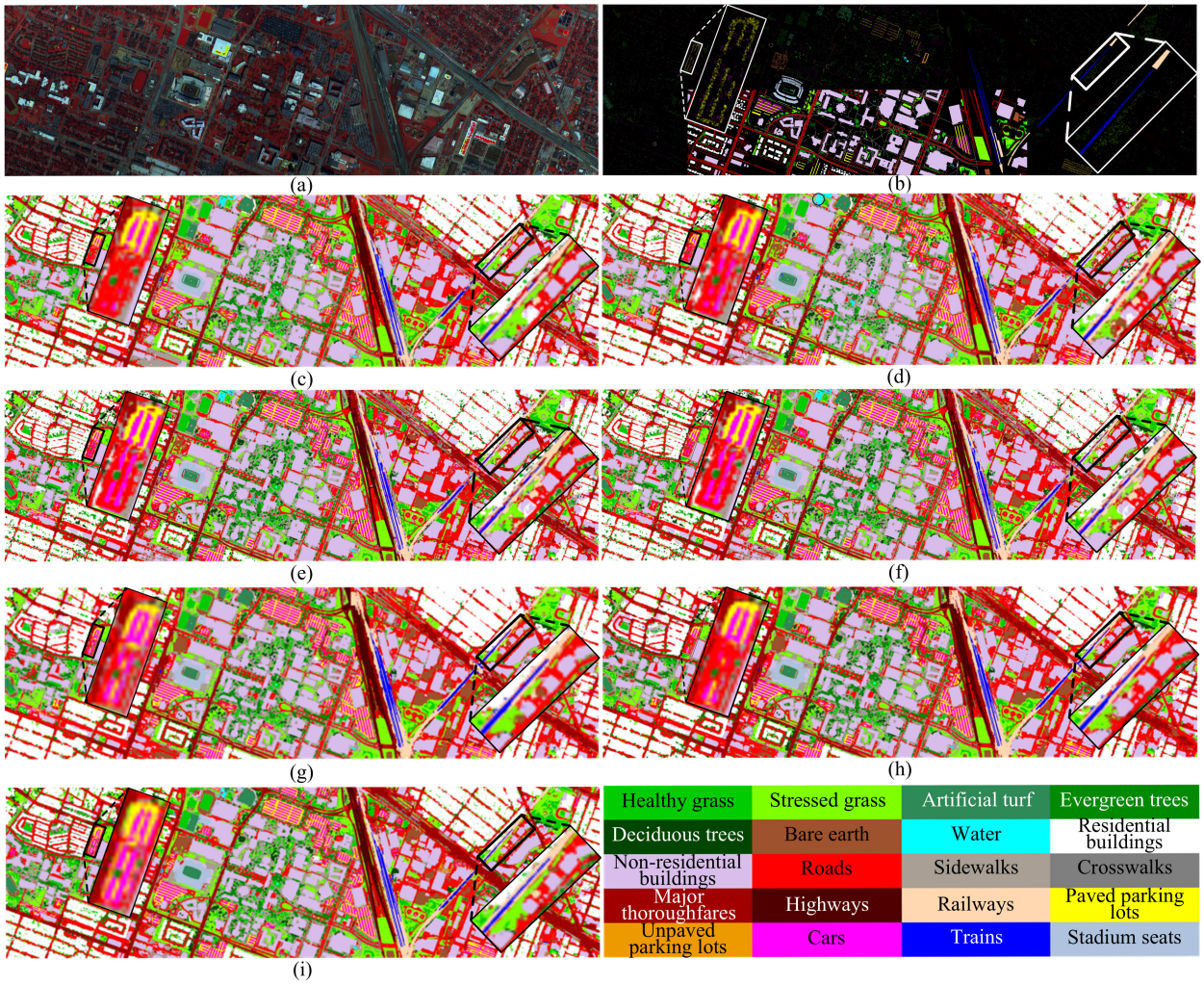


Fig. 8. Visual comparison on the DFC2018 dataset. (a) False color image. (b) GT. (c) CALC. (d) MFT. (e) Flex-MCFNet. (f) LoGoCAF. (g) CMX. (h) Sigma. (i) UniMamba.

TABLE VII
COMPARISON WITH RECENT MAMBA-BASED METHODS ON DIFFERENT DATASETS

Methods	Supported Modalities	DFC2013 (HSI-LiDAR)		DFC2018 (RGB-MS_LiDAR)		Hunan (MSI-SAR)	
		IoU (%)	OA (%)	IoU (%)	OA (%)	IoU (%)	OA (%)
RSGMamba	RGB-X	81.98	90.25	55.36	72.87	50.76	82.54
MSFMamba	HSI-LiDAR/SAR	83.27	92.24	50.02	69.78	48.08	81.72
MCAMamba	HSI-LiDAR/SAR	89.14	94.75	51.39	70.31	49.23	82.38
CoMiX	HSI-X	90.59	95.75	53.92	71.92	51.11	82.77
UniMamba	RGB-X / MSI-X / HSI-X	**92.48**	**95.86**	**57.03**	**75.05**	**52.89**	**83.26**

from efficient, modality-aware architectural design rather than excessive parameter expansion.

2) Ablation With HyperMamba, CroHMa, and FuseMa Variants: Table IX reports the ablation results for the HyperMamba, CroHMa, and FuseMa variants, with one component varied at a time while the others are fixed. Experiments assess the effects of forward and reverse spatial interactions, forward and reverse spectral interactions, and frequency-domain modeling.

Adding reverse spatial interaction (*Spa_b*) to the forward baseline (*Spa_f*) consistently improves performance, demonstrating the benefit of bidirectional spatial refinement. The forward spectral interaction (*Spe_f*) further improves OA across all three datasets. In contrast, reverse spectral interaction (*Spe_b*) generally reduces performance and provides only a marginal gain for HyperMamba on Hunan dataset. Fourier domain modeling (*Freq.*) further enhances feature representation, particularly in HyperMamba and CroHMa.

The best performance is achieved by combining bidirectional spatial interaction and forward spectral interaction in all three modules, together with frequency-domain modeling in HyperMamba and CroHMa.

B. Analysis of the CroHMa Module

1) Robustness to Geometric Misalignment: To evaluate CroHMa under geometric distortion, we conduct controlled

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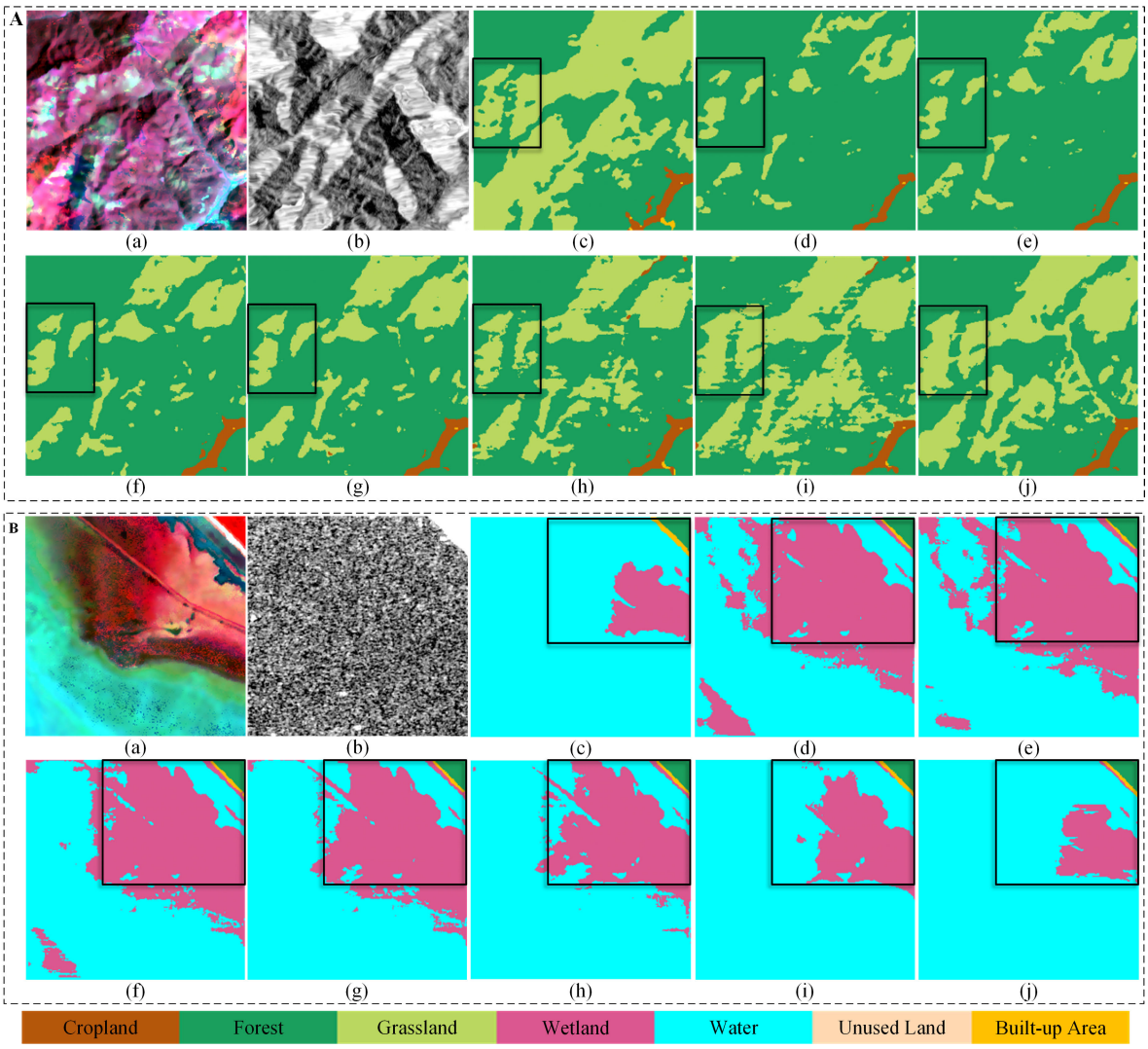


Fig. 9. Visual comparison on Hunan dataset. (a) False color image of Sentinel-2. (b) Sentinel-1. (c) GT. (d) CALC. (e) MFT. (f) Flex-MCFNet. (g) LoGoCAF. (h) CMX. (i) Sigma. (j) UniMamba.

TABLE VIII

ABLATION STUDIES ON THE HYPERMAMBA, CROHMA, AND FUSEMA MODULES. AVG. IS THE AVERAGE FUSION

Method		(a) Baseline	(b) HyperMamba	(c) CroHMa	(d) FuseMa	UniMamba
(a) Baseline		✓	✓	✓	✓	✓
(b) HyperMamba		✗	✓	✓	✓	✓
(c) CroHMa		✗	✗	✓	✗	✓
(d) FuseMa		*Avg.*	*Avg.*	*Avg.*	✓	✓
DFC2013	mIoU (%)	82.54	84.97	87.76	86.46	**92.48**
	OA (%)	88.30	90.83	93.48	92.37	**95.86**
	Params (M)	4.16	4.36	5.26	6.12	6.41
	FLOPs (G)	80.01	81.35	83.58	88.82	90.46
DFC2018	mIoU (%)	50.60	52.91	54.25	54.06	**57.03**
	OA (%)	69.01	71.08	73.21	72.87	**75.05**
Hunan	mIoU (%)	45.78	47.03	50.46	48.95	**52.89**
	OA (%)	79.21	80.74	82.50	81.60	**83.26**

spatial perturbation experiments on the DFC2013 dataset. Specifically, we compare UniMamba with its variant where CroHMa is replaced by cross attention. Both models are evaluated under simulated spatial shifts ($P \in \{0, 5, 15, 30\}$ pixels) applied to the auxiliary modality.

As shown in Fig. 10, CroHMa significantly outperforms the cross-attention variant under mild-to-moderate shifts ($P \leq 15$), demonstrating strong robustness to local spatial offsets. However, under severe distortion ($P = 30$), CroHMa degrades sharply and falls slightly below the cross-attention baseline. These results indicate that CroHMa is well suited

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TABLE IX

ABLATION STUDIES ON VARIANTS OF HYPERMAMBA, CROHMA, AND FUSEMA. SPA_F AND SPA_B DENOTE THE FORWARD AND REVERSE SPATIAL FEATURE INTERACTIONS, RESPECTIVELY; SPE_F AND SPE_B REPRESENT THE FORWARD AND REVERSE SPECTRAL FEATURE INTERACTIONS, RESPECTIVELY; AND FREQ. REFERS TO THE FOURIER DOMAIN FEATURE MODELING

Module	*Spa_f*	*Spa_b*	*Spe_f*	*Spe_b*	*Freq.*	OA (%) DFC2013	OA (%) DFC2018	OA (%) Hunan
Hyper_ Mamba	✓	✗	✗	✗	✗	89.56	70.63	79.25
	✓	✓	✗	✗	✗	92.56	73.64	81.15
	✓	✓	✓	✗	✗	94.79	74.82	82.84
	✓	✓	✓	✓	✗	94.68	73.56	82.89
	✓	✓	✓	✗	✓	**95.86**	**75.05**	**83.26**
CroHMa	✓	✗	✗	✗	✗	93.99	71.62	80.09
	✓	✓	✗	✗	✗	94.45	72.66	81.06
	✓	✓	✓	✗	✗	95.06	74.71	83.05
	✓	✓	✓	✓	✗	94.71	73.46	82.26
	✓	✓	✓	✗	✓	**95.86**	**75.05**	**83.26**
FuseMa	✓	✗	✗	✗	✗	93.89	73.36	81.96
	✓	✓	✗	✗	✗	95.36	74.56	82.78
	✓	✓	✓	✗	✗	**95.86**	**75.05**	**83.26**
	✓	✓	✓	✓	✗	95.28	74.18	81.69
	✓	✓	✓	✗	✓	94.76	74.26	82.36

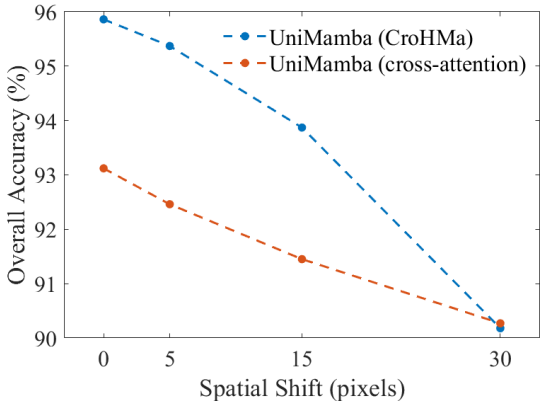


Fig. 10. Performance of CroHMa and cross attention under geometric misalignment.

TABLE X

EFFICIENCY COMPARISON OF CROHMA AND CROSS ATTENTION IN UNIMAMBA

Method	Params (M)	FLOPs (G)	Infer (s)
UniMamba (Cross-Attention)	7.01	95.89	17.36
UniMamba (CroHMa)	6.41	90.46	13.16

to relatively well-aligned multimodal data but remains limited under severe hardware-level misalignment or large geometric distortion, such as MSI-SAR fusion over rugged terrain. To mitigate this limitation, integrating deformable convolutions or adaptive spatial resampling before Mamba-based fusion represents a promising direction for future work.

2) Computational Efficiency Analysis: Table X compares the computational efficiency of UniMamba with that of its cross-attention variant under the same backbone setting. Replacing cross attention with CroHMa reduces the Params from 7.01M to 6.41M, FLOPs from 95.89G to 90.46G, and inference time from 17.36 to 13.16 s. These reductions are mainly attributed to the linear-complexity selective state-space mechanism. Despite incorporating an FFT-based frequency branch, CroHMa maintains lower overall computational overhead, demonstrating its efficiency for high-resolution multimodal segmentation.

TABLE XI

ANALYSIS OF THE DOWNSAMPLING LAYOUT IN TERMS OF OA (%) AND MIOU (%) ON THE DFC2013 DATASET

Stage 2	Stage 3	Stage 4	mIoU (%)	OA (%)
✓	✓	✓	72.46	79.35
✓	✓	✗	78.36	87.83
✓	✗	✗	92.48	95.86

TABLE XII

ANALYSIS OF DECODER ARCHITECTURES IN TERMS OF MIOU (%), OA (%), AND TRAINING AND INFERENCE TIMES (S) ON THE DFC2013 DATASET

Decoder	mIoU (%)	OA (%)	Train (s)	Infer (s)
UNet [12]	85.60	91.33	20808.68	15.09
UPerNet [52]	85.13	89.63	21563.20	15.46
UNetRes [53]	86.92	91.86	22500.41	15.61
Channel-Aware Mamba [3]	82.81	90.57	23797.30	27.04
ALL-MLP [45]	92.48	95.86	20526.22	13.16

TABLE XIII

SENSITIVITY ANALYSIS OF FREQUENCY-DOMAIN THRESHOLDS S_{LOW} AND S_{HIGH}

Dataset	(0.1, 0.9)	(0.1, 0.8)	(0.15, 0.85)	(0.2, 0.8)	(0.2, 0.9)	(0.25, 0.75)
DFC2013	95.26	95.21	95.34	**95.86**	95.43	94.98
DFC2018	74.61	74.76	74.86	**75.05**	74.56	74.08
Hunan	82.96	83.01	83.15	**83.26**	82.56	81.96

C. Analysis of UniMamba's Layout

1) Downsampling Layout Analysis: As shown in Table XI, we evaluated three downsampling layouts on the DFC2013 dataset: before Stages 2–4, before Stages 2 and 3, and before Stage 2 only. The results indicate that downsampling only before stage 2 achieves the best performance, whereas additional downsampling at later stages reduces accuracy. Although downsampling reduces computational cost, excessive spatial reduction can discard fine-grained details needed to identify small or complex structures, such as roads, sidewalks, and stadium seats.

2) Decoder Layout Analysis: Table XII reports the effects of different decoders on UniMamba's segmentation accuracy and computational efficiency on the DFC2013 dataset. The ALL-MLP decoder achieved the highest mIoU (92.48%) and OA (95.86%) while maintaining the fastest inference speed (13.16 s), outperforming UNet [12], UPerNet [52], UNetRes [53], and channel-aware Mamba decoders [3]. These results indicate that the ALL-MLP decoder provides the most favorable accuracy–efficiency tradeoff for UniMamba.

D. Hyperparameter Analysis

The frequency-domain modeling uses two thresholds, S_{low} and S_{high}. We evaluate their sensitivity across the three datasets using different threshold combinations. As shown in Table XIII, UniMamba maintains stable performance across the tested settings, indicating limited sensitivity to these thresholds. The setting $(S_{\text{low}}, S_{\text{high}}) = (0.2, 0.8)$ achieves the

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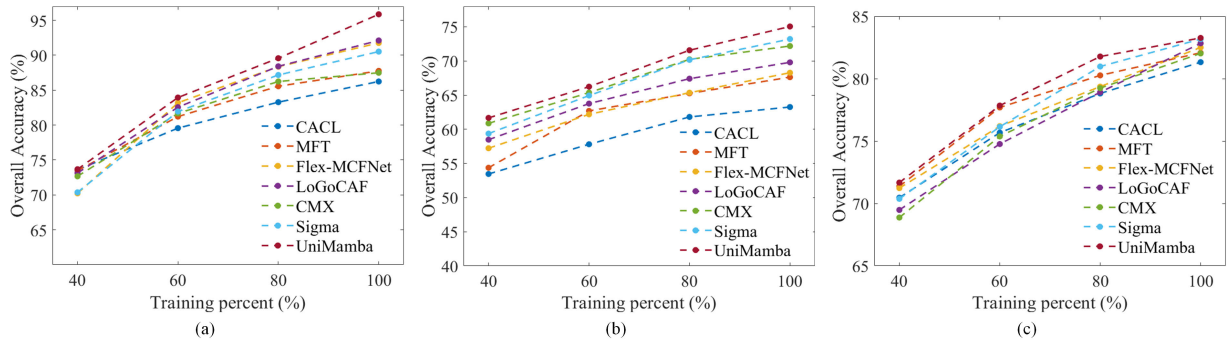


Fig. 11. OA versus different proportions of training samples for different methods. (a) DFC2013. (b) DFC2018. (c) Hunan.

highest OA and is therefore adopted as the default configuration.

E. Applicability and Generalization

We evaluated UniMamba under different training ratios (40%, 60%, 80%, and 100%) across the three datasets. As shown in Fig. 11, UniMamba consistently achieves the highest accuracy at all sampling levels, demonstrating strong data efficiency under limited annotation conditions. This performance benefits from its linear-complexity global context modeling, effective cross-modal fusion, and a lightweight decoder. The strong performance on Hunan dataset, which contains diverse terrains and complex real-world scenes, further indicates its practical applicability.

Nevertheless, these results do not establish cross-dataset generalization without retraining. In practical remote sensing applications, variations in sensors, atmospheric conditions, and geographic regions may introduce substantial distribution shifts. Moreover, the learned frequency-domain representations may be affected by dataset-specific frequency distributions when applied to unseen sensors or regions. Future work will investigate domain adaptation and self-supervised pretraining to improve cross-dataset generalization.

VI. Conclusion

This article proposes UniMamba, a unified Mamba-based architecture for cross-modal remote sensing semantic segmentation. HyperMamba jointly models spatial, spectral, and frequency-domain features to accommodate diverse 2-D and 3-D modalities. CroHMa enables efficient cross-modal interaction and implicit feature alignment, while FuseMa consolidates the enhanced modality-specific features into a unified representation. The lightweight ALL-MLP decoder aggregates multistage fused features for efficient segmentation prediction. Experiments on three benchmarks demonstrate the effectiveness and computational efficiency of UniMamba across RGB-X, MSI-X, and HSI-X settings. Future work will improve UniMamba's robustness to severe geometric misalignment and cross-dataset distribution shifts through adaptive spatial resampling, domain adaptation, and self-supervised pretraining.

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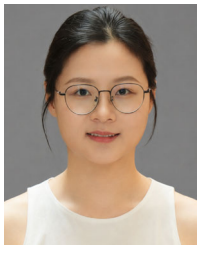
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